Quantum Teleportation

Quick Recap

• Superposition describes a qubit using a combination of basis states.

• CNOT is a controlled two-qubit gate.

• Entanglement links the description of two qubits in a shared quantum state.

1. What Is Quantum Teleportation?

Quantum teleportation is a protocol for transferring the state of a qubit to another qubit using an entangled pair, quantum operations and two classical bits. The original particle itself is not transported.

• Alice begins with the state she wants Bob to receive.

• Alice and Bob already share an entangled Bell pair.

• Alice measures her qubits and sends two ordinary bits to Bob.

• Bob applies corrections to recover the original quantum state.

Everyday analogy: Imagine transferring a secret recipe without moving the original sheet of paper. The information is recreated elsewhere, but the paper itself does not travel. This is only an analogy.

2. Why Can’t We Copy the Qubit?

An unknown quantum state cannot be copied perfectly. This is the no-cloning principle. Teleportation transfers the state instead of making a second identical copy.

• Alice’s measurement destroys the original state.

• Bob recovers the state after receiving the classical message.

• The protocol never leaves two perfect copies of the unknown state.

3. The Three Qubits and Required Resources

Qubit / resource	Role
Qubit 0 — unknown state	Alice’s state to transfer
Qubit 1 — Alice’s Bell qubit	One half of the entangled pair
Qubit 2 — Bob’s Bell qubit	Receives the teleported state
Two classical bits	Tell Bob which correction to apply

Alice: unknown qubit (q0) + Bell qubit (q1)

Shared entanglement: q1 ↔ q2

Bob: Bell qubit (q2) + Alice’s two-bit message

Figure 1. Three qubits and a two-bit classical communication channel.

4. Preparing the Bell Pair

Alice and Bob first create a shared Bell state, typically $|\Phi+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$. An H gate and then CNOT can prepare this pair from $|00\rangle$.

• The Bell pair is the quantum resource shared between Alice and Bob.

• Alice holds one qubit; Bob holds the other.

• Entanglement alone cannot finish teleportation.

5. The Teleportation Procedure

Step 1 — Prepare: Alice has the unknown qubit; Alice and Bob share a Bell pair.

Step 2 — Apply CNOT: Alice applies CNOT with her unknown qubit controlling her Bell qubit.

Step 3 — Apply H: Alice applies a Hadamard gate to the unknown qubit.

Step 4 — Measure: Alice measures her two qubits, obtaining two classical bits.

Step 5 — Send: Alice sends the two bits to Bob through an ordinary classical channel.

Step 6 — Correct: Bob applies the corresponding gate or gates and recovers the state.

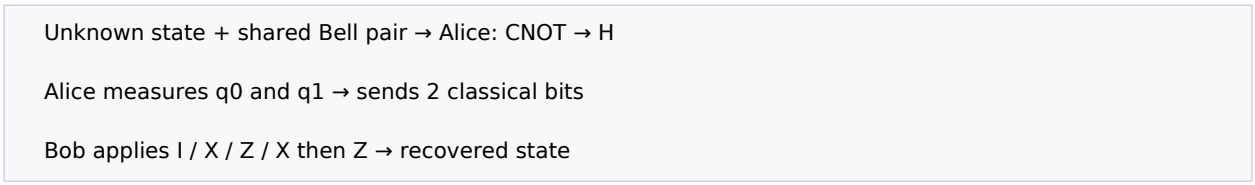


Figure 2. Teleportation workflow, from preparation to Bob’s correction.

6. Bob’s Correction Table

Alice measures	Bob applies
00	I (do nothing)
01	X
10	Z
11	X then Z

The two classical bits specify the correction. They do not directly describe the unknown amplitudes of the state.

7. Essential Mathematics

• Unknown state: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

• Normalization: $|\alpha|^2 + |\beta|^2 = 1$

• Shared Bell pair: $|\Phi+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$

• After Bob’s correction, his qubit is in the original state $|\psi\rangle$.

8. Worked Example

Suppose Alice prepares $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. She follows the teleportation steps and measures 10. From the table, Bob applies Z. After correction, Bob holds $|+\rangle$.

9. Quantum Circuit and Qiskit Example

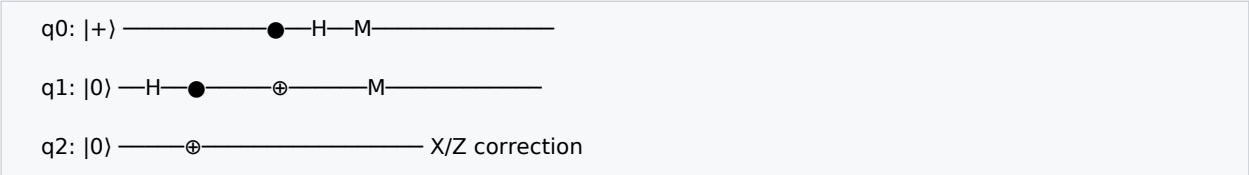


Figure 3. Conceptual three-qubit circuit; measurement bits control Bob’s correction.

```
from qiskit import QuantumCircuit, transpile
from qiskit_aer import AerSimulator

qc = QuantumCircuit(3, 2)
qc.h(0)            # State to transfer: |+>
qc.h(1)
qc.cx(1, 2)        # Shared Bell pair
qc.cx(0, 1)
qc.h(0)            # Alice's operations
qc.measure(0, 0)
qc.measure(1, 1)    # Alice measures
# Conditional X and Z corrections on Bob's q2
# (syntax depends on the Qiskit version)
```

The workbook uses conditional corrections with c_if, followed by AerSimulator execution and measurement counts. The key lesson is the circuit sequence and classical feed-forward. The code above is a conceptual excerpt, not a complete executable demonstration.

10. Applications

• Quantum networks — transfer states between distant quantum processors.

• Quantum repeaters — support longer-distance quantum communication.

• Distributed quantum computing — connect multiple quantum processors.

• Quantum internet and secure communication — future network building blocks.

11. Common Beginner Mistakes

“Particles travel instantly.” Only the quantum state is transferred.

“Teleportation is faster than light.” Bob must wait for two classical bits.

“The original state stays intact.” Alice’s measurement destroys it.

“Entanglement alone is enough.” Classical communication is also necessary.

“Teleportation copies a qubit.” It transfers without cloning.

Summary: What to Remember

• Teleportation transfers quantum information, not physical matter.

• Three qubits are involved: one unknown state and a shared Bell pair.

• Alice applies CNOT and H, then measures her two qubits.

• Alice sends two classical bits; Bob uses them to choose I, X, Z or X and Z.

• The original unknown state is not duplicated.

• Teleportation does not permit faster-than-light messaging.